

An Overview of Models for Business Process Analysis – Beyond Performance Measurement with KPIs

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Abstract—Almost all of today's performance measurement systems (PMS) rely on the usage of numeric indicators to quantify success or failure, normally referred to as key performance indicators (KPIs). This restriction makes it difficult to describe hard-measurable or business processes that are rather qualitative in nature. So, other existing models for process analysis containing convenient solutions and indicators have to be assessed and validated in order to offer them to stakeholders to thoroughly assess business processes. In this article, various existing models and frameworks for performance measurement were reviewed in an empirical investigation, categorized in systems and put into context to relevant theories. The result of the research is an overview of models for business process analysis that also incorporates indirect process performance aspects like soft goals, complexity, maturity or dependencies.

Keywords—PMS, Process Performance, KPI, Success Factors, Ontologies

I. INTRODUCTION

The state-of-the-art instrument to assess process success or performance is the use of numeric key performance indicators (KPIs) since they make a process measurable and controllable. But when KPIs are used as sole source model for process success control and performance enhancement, there's a certain danger that other relevant or unique features of a process might be not recognized and lost for review and control. This will be the case when processes or their results are non-measurable and can better be described in models with indirect indicators like success factors, soft goals, complexity, maturity, relations or dependencies. These indicators and models containing them must not be neglected because they might be a better way of understanding a process performance problem. The goal of this contribution is to offer an overview of today's used relevant and implementable models for business process analysis explicitly including models with indirect indicators for process performance.

Taking additional indirect indicators and their models into consideration for performance assessments will consume resources. For above mentioned reasons and from an organizational standpoint, it is preferable to gain the complete overview of a business process' performance to thoroughly understand, reengineer and manage it. In contrary to that, no such overview solution exists yet. Enterprises and researchers, the stakeholders of this research would be forced

to browse through many possible matching models in order to assess and validate them for their specific process performance problems. The overview is designed to facilitate this search to some extent.

Proceeding with the description of the state of the art, then the used research design is proposed. Finally, the presentation of the investigation course and the results frame this paper.

II. STATE OF THE ART

There's a multitude of models for business process analysis as well as performance measurement systems available on the market and being discussed in research. Since the terms performance and success in combination with measurement are used synonymously in most publications, objective and scope of proposed systems are not obvious and have to be reviewed in detail. Two different approaches have anyway to be differentiated: Performance measurement systems, i.e. frameworks that describe how the performance of a business process is measured and models, i.e. the inherent concept of way the processes are working and can be assessed. At least, they have in common to give stakeholders a guiding principle to review or assess processes and contain or use more or less suitable indicators that can be used for assessment.

A. Performance Measurement Systems

Most framework models, normally referred to as performance measurement systems (PMS) exist that display measurables in a certain overview or dashboard style; most prominent is the concept of the Balanced Scorecard (BSC, [10]). It discloses in particular only numeric parameters from four different sources: financial, customer, internal business processes as well as learning and growth. It also combines these measurables with strategic goals and is used with a usual review frequency of no less than a month [1]. So basically, BSCs display parameters that were generated elsewhere; it is no generic model for process analysis actually generating KPIs. Other PMS using numeric indicators similarly and exclusively are amongst others:

Performance prism [30], the SMART/performance pyramid [31], EFQM [21], the performance management questionnaire [32], supportive performance measures/performance measurement matrix [33], results/determinants matrix [34], intangible success factors [35], dynamic performance measurement system [36], Sink and Tuttle model [37], integrated dynamic performance measurement

system, [38], Scandia's intellectual capital navigator [39] or CMMI [19]. Some other approaches are cited as PMS, but turn out to be assessment methods and tools for evaluating measures rather than frameworks when drilling down into full text, e.g. Neely's assessment and measures development [8] or workflow-based monitoring, self-assessment, benchmarking and statistical process control, all amended in [14].

B. Key Performance Indicators

Out of the investigation of the PMS, by far most of the prevailing models recommend the usage of numeric KPIs in diverse specifications to monitor and control processes according the business process management approach [5], hence to make the process success visible and measurable. In addition to that, KPIs are often used exclusively in specific functional domains like production, IT or supply chain and deal with the processes' efficiency. This might indicate that numeric models concentrate on processes with value creation and disregard accompanying processes like finance or R&D that *only support*, but also in service domains of organizations, KPIs begin to spread. In a McKinsey study [4], IT departments used productivity, timeliness and quality as parameters, thus being borrowed from classical KPIs for production. Apart from productivity parameters, financial KPIs like the EBIT (earnings before interest and tax), or liquidity exist for a long time. In another study, 51% of all enterprises mentioned the use of financial besides productivity and quality indicators [24]. In fact, some more sources ([6]; [22]) claim the display, combination, weighing and balancing of financial as well as production KPIs as one of the most important usages of a KPI system.

Moreover, there are so-called SD-KPIs (sustainable development-KPIs). By german law, suitable related indicators have to be disclosed as soon as they are of importance for the understanding of the Business development, result or the situation [3], e.g. environmental or labor indicators. They are primarily used to display a certain overall responsibility and cannot be broken down into department or individual performance values like classical KPIs. Though, they serve the assessment paradigm and try to make enterprises comparable, at least within a given branch or timeframe.

C. Other Models

However, there are some more models that suggest verbal descriptions as indicators of success. In contrast to that, few approaches exist that question and assess processes in their overall structural effectiveness and design; they are embodied by maturity models and Ontologies for business processes. Moreover, the only PMS that could be identified as not exclusively using numeric indicators was the process performance measurement system, introduced by [14]. It also considers qualitative aspects of performance through the definition of goals and recommends the usage of indicators that state to what extent the goals are fulfilled.

For overview reasons, the other models and indicators are discussed in detail in the investigation course part of this paper, since their contents combined with KPIs actually

frame the overview solution. During the research so far, no other overview construction about non-numeric indicators or models for business process performance could be found.

III. RESEARCH DESIGN

A five-stage design approach is chosen to conceptualize an artifact that enables stakeholders to select from a multitude of relevant models for business process analysis. After the presentation of the overall research scope in a semantic map, the chosen design science approach is explained. Then, relevant theories of information science are reviewed to give answers what a business process or, derived from that, a model to analyze it should look like. After this discussion, the method of evaluating data is presented. Ultimately, the research questions are postulated. Using this research design, the investigation course took place.

A. Overall Research Scope

Originating from the problem for stakeholders to gain a comprehensive overview or to select a suitable model for their needs, the overall research scope can be formulated and separated into two branches.

First, the review of known models reveals that there are solutions for single process domains and problems as well as some combined models for multiple domains (like PMS), but no solution exists that displays a broad range of multiple available models. From this *model system* approach the necessity to evaluate either an overview, combined, or integrated solution is derived. The structural framework of the proposed overview solution embodies this goal in a first step. It offers quick access and overview but still only refers to many models to choose from. So other framework or integration possibilities have to be evaluated in the future to further facilitate the implementation of performance assessment possibilities in business processes. Ultimately, a one-step model that combines or integrates the various models should be deployed in further work.

Second, stakeholders must perform research or consult experts when looking for a suitable model. This and the validation of the specific applicability would be very time-consuming. The empirical research in this study, combined with a review of relevant theories on this *model design* approach side delivers data to describe the final overview solution by means of designing an artifact according to the design science idea discussed in the following section. The overall approach and its two branches again are visualized in a semantic map at the end of this paper (Figure 1). Parts of the map that are covered by this contribution are printed in white copy on tint.

B. Design Science

Design science as the approach of choice has its roots in engineering science and is an iterative, modeling, prototyping and focused on problem solving way to build theories. Its results are specially designed artifacts which are primarily created to solve tangible problems of a company. In comparison to other scientific disciplines in which common best practice approaches of research are basic knowledge, but the research results might be questionable

and subject of interpretation, design science oriented researchers nowadays face a certain discord about commonly used clear and accepted guidelines for their building approaches and findings. Depending on the various used methods and processes as well as their standing as scientific, rigorous, qualitative, and relevant, the embodiment of artifacts will vary as well. In contrary, the acceptance and validity of artifacts themselves as relevant objects of research is not doubted.

The concept of an artifact is defined in different ways. Hevner et al. [15] analyzed design theories and categorized their artifacts by their form into constructs (vocabulary and symbols), models, methods and instantiations (i.e. material systems). Gregor and Jones [12] differentiate by the purpose of the artifacts. They categorize by instantiation (concrete artifacts) either as a product or method as well by theories (abstract artifacts) which describe the inherent principles in the design of the artifacts. According to their work, a design theory (a finding about an artifact in terms of design science, therefore a research contribution) should be considered as described sufficiently and as relevant in case of the contents of the design theory can be adequately described in following eight categories: purpose and scope, constructs, principle of form and function, artifact mutability, testable propositions, justificatory knowledge, principles of implementation and expository. Iterative review of all of these criteria's presence is necessary and desirable to reveal fundamental gaps. In the present contribution, the guidelines of Gregor's work shall be kept in mind to describe the design theory for a overview of models for business process analysis, which is the expected artifact proposed in this work.

C. Relevant Theories

According to Porter's five forces, a successful company can enhance its competitive situation by influencing the balance of power in its market situation by five aspects: branch internal rivalry, buyer power, supplier power, potential substitutors' power and market entry constraints [23]. In this context, Porter's theory can be applied in three ways to companies who investigate their business processes:

- The mere knowledge of the theory can be a trigger for the company to start moving and question its competitiveness. Without explicitly applying the theory itself, basic process improvement instruments, e.g. the usage of KPIs drive direct and indirect cost reductions as well as a reduction of market entry barriers.
- Applying the theory reveals hidden relations between processes, their building blocks and so highlight entry points for Porter's forces. For example, a process might be detected as unquestionable supplier-orientated. So, the processes become more transparent.
- Finally, hidden rivalries and strategies between departments and divisions become obvious when reviewing similar or equivalent processes. This approach emphasizes the comparability of processes.

The last aspect is being discussed in the principal-agent-theory in particular. It postulates the existence of a lack of information necessary to make decisions in a principal-agent relationship. On the one hand, this comes from information held back for reasons of personal advantage, on the other hand because of the accessible information in most cases being incomplete and unevenly distributed. To overcome this unfairness, so-called agency cost is generated [17]. There are three types of agency cost:

- Cost that represent the principal's work to reduce its information lack, called monitoring cost. They occur when the agent delivers too few information and that the principal has to gather additional data. Hence, the principal lacks transparency in the results of the agent's processes.
- Starting new relationships to a principal possibly comes with certain bonding expenditures to the agent. To avoid information dissymmetry, they are necessary for the agent to signal its information to the principal as transparent and useful.
- Residual cost occur whenever the distribution of necessary information is suboptimal, e.g. if a time-consuming workaround has to be established to gain relevant information out of standard software formats. This aspect emphasizes a certain inhomogeneity of necessary data and thus incomparability.

According to the organizational information processing theory, enterprises are not only creating potentially obsolete buffers to balance supply shortages in production but also to catch uncertainties and information shortages in every part of the company [25].

- The more independent regional, temporal or organizational entities exist in business processes, the more uncertainties are observable, e.g. the existence of multiple repositories for identical documents in different departments because they may are not be working transparent to others.
- Moreover, the theory claims that enterprises tend to avoid these uncertainties at the same time by the use of adequate tools. They should be made available to others as well to design their processes in a transparent and comparable way.

Summing up, all these theories target to make a business and hence it's processes more transparent and comparable irrespective of their functional domains. Subsequently, results, constraints and properties of processes will be more visible. All in all, there is a necessity to examine if models for business processes work towards this needed transparency or comparability in principle.

D. Empirical Investigation

From the initial standpoint of an exemplary enterprise seeking advice in selecting a suitable model for its business process performance problem, the currently available

management approaches, commercial offers and scientific models were evaluated in an empirical investigation.

About 30 publications of each public available models and PMS with accompanying comparisons or reviews were collected and reviewed from March 2009 to March 2010 according to a systematic review scheme [7]. The models especially were selected in accordance to the complementing requirement to satisfy the needs of companies with a certain size, process depth and diversified organizational functions like marketing, supply chain or finance. Enterprises of this kind, in other words potential stakeholders of this research most likely will be sensitized for or even familiar with concepts of quality systems, Reengineering or BPM and the implementation of models like KPIs. Moreover, broadening the business process conceptualization view from industry to adjacent service or administration and research sectors revealed further interesting approaches and methods, so that models with backgrounds in these directions were also taken into consideration. From that point of view, first of all the relevant standards, norms and statistical methods had to be reviewed. Since enterprises themselves will potentially not disclose any or very few information and insights in this delicate domain of doing business, relevant data had to be collected mainly from secondary sources. This was done via the analysis of publications of inter-trade-organizations, branch reports of accountants, examining institutions or other stakeholders. Apart from the valuable facts, they also allowed a deeper look into the originators' various approaches of analyzing processes. All in all, about 15 models that could be used at least in this range of enterprises remained for detailed investigation, so did 10 distinctive PMS.

E. Research Questions

- Can other models and systems be found that do not use KPIs exclusively?
- Can they be categorized in a meaningful way and so, can an overview table be created?
- Will the overview solution enable enterprises and researchers to better select and compare models for business process analysis?
- To what extent does the artifact fulfill the basic business process requirements from the relevant theories?
- Is the overview easily applicable, extendable or mutable?
- What constraints are visible using the overview?
- Can it be seen as a basis for further investigations?
- What other perspectives are visible?

IV. INVESTIGATION COURSE

To categorize the models, a certain distinctive feature has to be defined. In this context, the relevant *model* for business process analysis shall allow

- to describe the process analysis and assessment approach to a certain meaningful extend
- to perceive the domain of its validity as well as to assess its performance with a meaningful indicator.

As the performance of a process does neither has to be measurable nor numeric by default, the concept of the generic *indicator* has to be defined and is being used in this work. An indicator can be

- a numeric parameter that evaluates the process (puts a certain value to the performance) or
- a verbal description (factor) that rates the process (describes and judges the performance).
- It is used to assess process performance by means of
- process success (indicating efficiency) or a
- process outcome (indicating effectiveness) respectively.

TABLE I. OVERVIEW OF PROCESS PERFORMANCE SYSTEMS AND INDICATORS

Performance Assessment	Process Success / Efficiency		Process Outcome / Effectiveness	
Indicator	Parameters	Factor	Parameters	Factor

Browsing through the reviewed models, it can be observed that they all can be categorized into these four process performance systems dependent on the occurrence and application of their indicators.

A. Parameters for Process Efficiency

The widespread concept of lean manufacturing enables enterprises to identify and eliminate waste in processes with a clear focus on countable results (measured via KPIs) in value creating processes. Outside of the boundaries of manufacturing or value creation, processes usually are not taken into consideration by the lean concept. The six-sigma approach qualifies process success which is described with critical to quality characteristics (CTQs). It also is limited to value creation processes with verifiable lacks displayed within these indicators [28].

TABLE II. EXAMPLES OF PARAMETERS FOR PROCESS EFFICIENCY

Parameters	
Amount	Output, Cash, Amount of new Clients per day
Time	Cycle Time, Mean Time between Failure (MTBF), Inventory Turnover
Quality	Scrap Rate, Variance

B. Factors for Process Efficiency

Factors for process efficiency rather rate the process success verbally and abstract than evaluating it with a figure. The factors can take shape of either a binary yes-no decision or a discrete cascade of opinions about the process success. The opinion cascade can be a set of verbally escalating statements or quantized quasi-values that embody a success rate from virtually 0 to 100%.

Performance management, i.e. tracking the achievement of an employee's goals is a common way of developing staff in enterprises and a good example for process success factors. They normally form verbal goals that are rated in an opinion cascade as a failure or success. A similar approach is postulated by Zellner [29] in CRM. He suggests claiming a success whenever an abstract goal in a certain phase of the CRM has been archived, repeated throughout the whole customer lifecycle.

The European foundation for quality management (EFQM) offers an excellence-model for enterprises that rates multiple so-called enablers and results by relevance, performance, approach, application, assessment and improvement [21]. A maximum of one thousand points can be granted for the total of these criteria. So, the EFQM model also offers a quasi-value for the success of a process (in this case the complete company).

Huth offers a six-stage assessment model in his work about the comparison of Lean and Six Sigma principles that also can be used to rate all other processes' success [16]. He suggests using various criteria in the categories strategy, approach, methods, culture (implementation) and impact that can be subject to an either-or decision or certain weighting.

TABLE III. EXAMPLES OF FACTORS FOR PROCESS EFFICIENCY

Factors	Yes-No-Decision	Opinion Cascade
Marketing	Kept Customer Base Started further negotiation	Customer Satisfaction (0-100%)
Human Resources	Implemented Talent Development Program Participated in Graduate Fair	Incentive System implemented (rudimentally, partly, fully)
Research & Development	Expanded Alliances with key suppliers New Product Market-Readiness finished	Perceived Renommé (0 to 5 Points)

C. Parameters for Process Effectiveness

Processes can be reduced to their basic building blocks and immanent parameters which describe them in detail as a system with the function to execute a certain task. This reveals the existence of many collective numeric indicators which are independent of complexity or maturity and are able to assess a process in total or its pieces. They can be quantitative like instances, dwell, and departments or qualitative like granularity or necessary knowledge. Though, most of the commonly used models to disclose process effectiveness or outcome in industry show characteristics of maturity models.

A simple five-stage set of values is being proposed by Rummler et al. [26] in their process maturity model and process performance index (PMM/PPI). By far more

complex is the capability maturity model integration (CMMI) methodology that originally has been proposed for IT projects [19]. It breaks down the scope of the assessment into process areas and rates their maturity in five levels. In addition, it suggests the use of metrics and indicators in certain process areas, which again emphasizes the problem that terms as well as content of definitions like process, project and their success and outcome must be clearly separated and not be mixed up.

The approach of formulating goals and their optimum parameters in a specific language (goal driven language, GDL) to automatically optimize business processes in a business process execution language (BPEL) algorithm is proposed by Kramberg [20].

Hammer's process and enterprise maturity model (PEM) is a radical enterprise-independent process reengineering approach for practitioners and managers and has advanced to a certain state-of-the-art-Model over time [13]. He assesses the maturity of a company by claiming a success for specific factors of a verbally escalating four-level opinion cascade, both for the areas of process determinates and enterprise competencies.

TABLE IV. EXAMPLES OF PARAMETERS FOR PROCESS EFFECTIVENESS

Parameter	Quantitative	Qualitative
Amount	Steps (Instances), Documents	Granularity of data to be processed Amount of various Scenarios
Time	Cycle Time, Deadline	Man hours in Full time equivalents Response Time
Relevance	Departments involved, Stakeholders	Severity of task to be processed, Maturity

D. Anomaly RACI

In project management, for role declaration and assignment, the RACI model is often used. It allocates the roles responsible, accountable, consulted and informed to the actors [2]. Many sub-models exist that include further roles like omitted or support. The roles are usually only associated to names when a specific project, an instance of a process is being started.

The model describes the relevance of roles. Hence, it can be considered as description of how the business process works through its actors, and as an attribute of the effectiveness of the process. It is non-numeric and attaches a certain attribute to a part of business process, like in a taxonomy or an ontology.

E. Factors for Process Effectiveness

Another possibility to describe processes of any kind are ontologies. In information science, they embody formal specifications of a concept [11]. In business, they are used often in the context of knowledge management and expert systems, but not very frequently to describe what a business is doing. They're able to classify and describe a concept's (in this case processes') instances, properties and relations. Suitable analyzing programs (so-called reasoners) allow users to ask qualified questions to an ontology. The answers will not contain crude data like a database extract but may be

a meaningful statement on the same semantic level as the question, either as binary yes-no statement or as an open description. The answers to a question to an ontology can hence be considered as a description of the result and outcome of the process. In addition to that, the answer can be rated as more or less meaningful and sufficient as a solution for a problem. Thus, qualified questions to ontologies can be interpreted as success factors for the detection of process effectiveness to some extent, and in this respect, rated answers from ontologies can be taken into consideration to be their actual indicators, factors for process effectiveness.

Gehre et al. (Project Intelligrid) offer a catalog of a business process-(domain knowledge), service-, organizational- and software resources Ontologies for the use in virtual, distributed workgroups like in project management [9].

Jenz [18] recommends a three-stage ontology model for business processes, namely a basic business process management ontology (BPMO), an industry-specific- as well as an organization specific ontology. He proposes to combine them into a single business ontology and technical representations, e.g. an IT-ontology. This historical approach of describing and somehow condensing a domain into a hierarchy of several ontologies was extended in recent research. Based on the approach of a service oriented architecture (SOA) in which context ontologies can be seen as independent services to be coordinated, the specific Ontologies with domain knowledge are supported by choreographic ontologies which contain the interfaces and translating elements.

The team NESSI offers the development of a BPM software for cross-domain modeling of processes. It is designed to use already existing elements of object- or workflow-oriented programs and semantic frameworks (ontologies). It also differentiates between domain specific and choreographic concepts [27].

TABLE V. EXAMPLES OF FACTORS FOR PROCESS EFFECTIVENESS

Qualified Questions	Yes-No-Question	Open Question
Production	Is the continuous production endangered in case the Shift Manager is sick?	In which Machine Family Training efforts are lowest?
Finance	Does the new law affect our product costs?	What cost impact has it to drop a certain Process Step?
Human Ressources	Do I have to hire extra staff for the new Process?	What minimal qualification is necessary for the processor at this task?

V. FINDINGS

The following table combines the domains, models and indicators of the previously assessed models into an overview of models for business process analysis. All in all, the table displays to what extent models are applicable in a certain domain as well as for the assessment of a specific process performance problem, either fully (X) or partly (O).

TABLE VI. OVERVIEW OF MODELS FOR BUSINESS PROCESS ANALYSIS

Models	Efficiency		Effectiveness		Process Domain
Indicator	Parameter	Factor	Parameter	Factor	
Lean Management	X				Value Adding Processes
Six Sigma	X	O			Value Adding Processes with Quality Problems
Performance Management		X			Human Ressources
CRM		X		O	Customer Relationship
EFQM		X		O	All Processes
Lean Six Sigma		O	X		Value Adding Processes
CMMI	O		X		IT-Projekte
GDL			X	O	All Processes
PEM/PPI			O	X	All Processes
RACI				X	All Processes
Intelligrid				X	Virtual organisations
BPMO				X	All Processes, domain specific
NESSI				X	All Processes, domain specific and choreographic

VI. CONCLUSION AND OUTLOOK

Models for business process analysis can be categorized into four process performance systems according to the occurrence and applicability of an individual indicator. Each indicator tries to disclose process performance in order to solve a performance problem. The models can be cataloged into an overview table which enables stakeholders to quickly identify and select a suitable model for a certain scope.

According to the claim of the relevant theories, a business and therefore its processes have to be transparent and comparable irrespective of its domain. From that point of view it can be stated that the models and the overview allow to enhance the transparency of the business processes via the usage of suitable indicators. The feature of comparability of processes and the models in the overview is still underrepresented because the models and indicators are mostly exclusive to a performance system or valid in most different domains and so, results cannot easily be exchanged and shared.

Picking up that point, the design of the overview also reveals that there are voids in the representation of models for more than one isolated performance problem. If an enterprise wants to thoroughly reengineer its processes, it should be able to evaluate their efficiency and to rate their effectiveness in any given business process without restriction to a specific process domain or performance

system. It should not be forced to use many different and isolated models. For both above reasons, it would also be preferable to find a more convenient overview solution. First, it should enable stakeholders to access the domain of scope easily, second, it should allow to evaluate the possible models in reach.

Moreover, by now, the existence of a multitude of models and ecosystems and lack of a certain comprehensive approach leads to the suggestion that a super ordinate or integrated model to cover the performance systems' problems today is lacking and should be deployed. This will also be subject to future research as presented in the overall research scope. The concept of the models must not be mixed up with PMS which are frameworks for performance measurement, but basically working with numeric parameters. It will also be subject to further research in the meantime to investigate, to what extent they already can be considered as combined or integrated models

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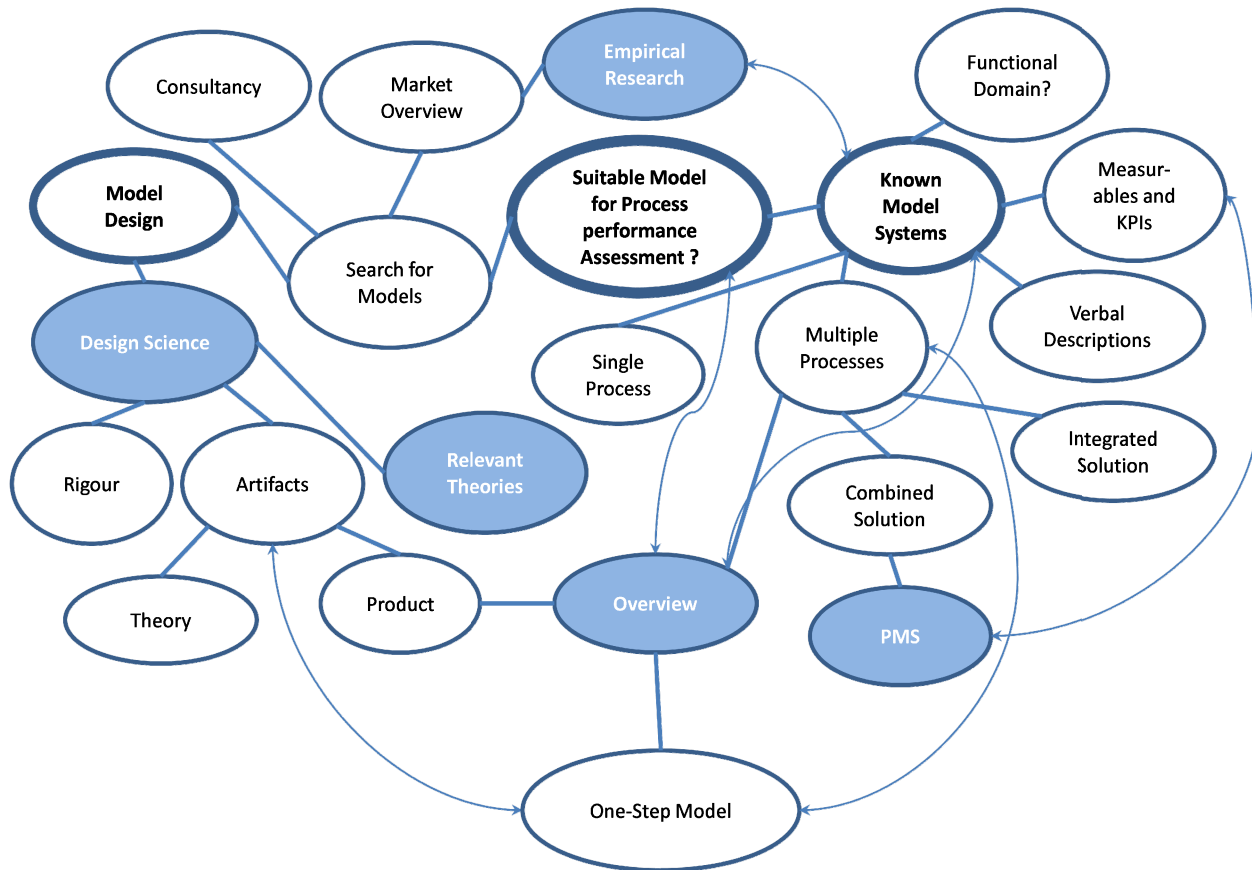


Figure 1. Semantic Map of Overall Research Scope